Korea Technology and Information Promotion Agency for SMEs

Technology Proposal for 2026 TIPA* International Joint R&D Program

(TIPA: Korea Technology and Information Promotion Agency for SMEs)

The objective of this Technology Proposal is to introduce the institute's proprietary technologies and present potential collaborative areas for Korean SMEs, facilitating joint research and development projects under the 2026 TIPA International Joint R&D Program. This Technology Proposal will be publicly disclosed in conjunction with the official announcement of the 2026 TIPA International Joint R&D Program.

1. Technology Proposer Information

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2. Technology Overview

Title of Proposed Technology	*Miniaturized, Cost-Effective and High-Performance Bio-Medical Joined Communication and Radar Sensing System for Autonomous, Contactless, Continuous and Real-Time Vital Signs Monitoring and Wireless Data Transmission (Bio-Medical JCAS System)*			
Korea's 12 National Strategic Technology Fields	Semiconductor/Display ( ), Secondary Battery ( ), Advanced Mobility ( ), Next-gen Nuclear Power ( ), Advanced Bio (X), Aerospace/Marine ( ), Hydrogen ( ), Cybersecurity ( ), AI ( ), Next-gen Communication ( ), Advanced Robotics & Manufacturing ( ), Quantum ( )			
Industrial Technology Classification [1]	**Technology Field 1** (Large Category)		**Technology Field 2** (Large Category)	
	Bio/Medical		*Mechanical / Materials*	
	Medium Category	Small Category	Medium Category	Small Category
	Therapeutic Devices and Diagnostic Devices	*Bio-signal Measurements / Diagnostic Devices*	*Automotive/Rail Vehicles*	*Safety Improvement Technologies*
	Functional Restoration / Assistive & Welfare Devices	*Daily Living Support Devices and Systems*		
	Medical Information & Systems	*Remote and Home Health Care Devices*		

Concept of Proposed Technology	• *Development, test and demonstration of a miniaturized, cost-effective and high-performance bio-medical joined communication and radar sensing system for autonomous, contactless, continuous and real-time vital signs monitoring and wireless transmission of the measured data.* • *Co-design, integration, and optimization of the components of radar and communication sub-systems into a single, miniaturized, and cost-effective hardware platform.* • *Radar sub-system is designed to operate at millimeter-wave band (to enable high resolution) and communication sub-system is designed to operate at lower frequencies (e.g., in the 6G - FR3 band) to enable long distance communication. Since radar and communication systems operate at different frequency bands, self-interference is minimized.* • *High-resolution radar sub-system is designed to accurately measure very small chest movements of target person(s) caused by vital signs (e.g. breathing, heartbeat). Based on the measurement of the reflected signal, advanced signal processing algorithms are applied to extract vital signs.* • *AI and machine learning based analysis of the vital signs are implemented for detection of psychological, heart and respiratory related disorders, stress, sleep disorder as well as for fall detection.* • *The measured vital signs are wirelessly transmitted using the integrated communication sub-system. This would then enable telemedicine and remote monitoring.* • *The miniaturized JCAS hardware system will consist of an integrated functional block for radar and communication signals operating at radio frequency (RF) and baseband functional block for digital signal analysis.* • *Off-the-shelf Microwave monolithic integrated circuits (MMICs) will be used. These ICs (chips) will be assembled using advanced packaging and system-integration technologies. The communication and radar antennas as well as the passive RF components (e.g., filters) will be designed, fabricated, and tested during the projects. Furthermore, a system-integration board for system integration of communication and radar sub-systems will be designed, fabricated, and tested.* • *Tests will be performed at components, sub-system, and system level.* • *The software components required for the functionality of the bio-medical JCAS system as well as for signal processing to extract the vital signs and for analysis of the vital signs will be developed during the project.* *Potential use cases* • *In-cabin sensing of drivers and passengers in cars / trains to detect medical conditions and detect presence and signaling to wireless car exterior networks* • *Contactless monitoring of patients in critical conditions in hospital environments and wireless ad-hoc transmission of measured data to available networks* • *Early prediction of heart, respiratory and psychological-related disorders using measured*

	vital signs, together with advanced AI and machine learning software algorithms • *Contactless sleep monitoring for point-of-care applications at home and wireless integration into medical data management* • *Contactless fall detection and wireless alarm signaling for handicapped and unconscious persons*
Rationale for Joint R&D Project	• *Bio-medical JCAS system consists of complex electronics hardware for RF, data acquisition, digital signal processing and signal analysis software* • *Development of complete bio-medical JCAS system requires extensive knowledge in electronic microsystems packaging, system design, RF engineering, signal processing and medical vital sign analysis* • *Know-how for all domains is rarely available at one institution / company* • *Collaboration allows to co-optimize the individual parts and the overall system* • *International collaboration encourages development for multiple country specific requirements of international target markets and thus enables wider exploitation* • *A successful joint R&D collaboration would enable an innovative bio-medical JCAS system with the following advantages* 1. *Non-contact health monitoring: Reduces risk of infection and improves comfort for patients, especially in hospitals, homes, and elderly care.* 2. *Continuous, real-time data: Enables early detection of health issues (cardiac/respiratory problems, sleep disorders, stress) and timely interventions.* 3. *Integrated solution: Combines sensing and communication in one miniaturized hardware, lowering cost, power consumption, and device footprint versus separate systems.* 4. *Improved accessibility: Wireless data transmission supports remote supervision, emergency alerts, and scalable care models.* 5. *Enhanced safety and autonomy: Autonomous data collection and transmission reduce the need for clinician presence and enable continuous monitoring in remote or underserved settings.*
Key Words	1. *Contactless vital sign monitoring* 2. *Wireless transmission of measured data using bio-medical JCAS system* 3. *Telemedicinal application for psychological and cardio-vascular related diagnosis/indications* 4. *Safety improvement for automobile and rail traffic* 5. *Sleep monitoring and fall detection*

3. Research and Development Objectives & Plan

Technology Readiness Level (TRL)[2]	Start of the Project	Level (3-4)	End of the Project	Level (5-6)
R&D Objectives	*The R&D objectives of this project focus on the development, test and demonstration of a miniaturized, cost-effective and high-performance bio-medical joined communication and radar sensing system for autonomous, contactless, continuous and real-time vital signs monitoring and wireless transmission of the measured data. Specifically, the R&D objectives include the following:* • *Development of system-level concept for integration of radar and wireless communication sub-systems into a single hardware platform* • *Investigation of parasitic interaction/coupling problems at system and component levels* • *Development of RF design measures for minimization/elimination of the interaction/coupling at components and system levels* • *Development of RF building blocks of radar and communication sub-systems using off-the-shelf MMICs* • *Evaluation, test and demonstration of building blocks and sub-systems functionality* • *Integration of building-blocks into bio-medical JCAS medical radar system* • *Development, test and implementation of radar signal processing algorithms for extraction of vital signs, considering single and multiple targets* • *Development, test and implementation of AI and machine learning based algorithms for analysis of extracted vital signs and use for prediction of heart, respiratory and psychologically related disorders* • *Development and implementation of suitable bio-medical JCAS communication protocol* • *Development and demonstration of bio-medical JCAS system*			
R&D Target Specifications	• *Continuous and real-time detection of breathing rate and heart rate of humans for automotive in-cabin, rail car, hospital and nursing home use-cases* • *Measurement range 0.2 m to 5 m* • *Capable of MIMO (multiple input multiple output) functionality to monitor multiple targets* • *Wireless transmission connectivity to internet* • *Component integration for both radar sensing and wireless communication systems in same but multi layered stacked low-cost PCB arrangement* • *Based on commercially available MMICs and standard manufacturing technologies*			
R&D Plan	***Phase 1 (6 months)*** ***Key Task 1: Generalized use case definition and requirements analysis (6 months)*** • *Input: Generalized use-case descriptions* • *Output: Use-case definition for near-target and far-target generalized use-cases including requirements (at component, sub-system and system-level) for continuous real-time bio-medical JCAS systems* ***Key Task 2: Preliminary investigation of hardware concepts (4 months)*** • *Input: Preliminary use-case definition and requirements (key task 1)* • *Output: Preliminary hardware concepts for near-target and far-target use-cases* ***Key Task 3: Preliminary investigation of software concepts (4 months)*** • *Input: Preliminary use-case definition and requirements (key task 1)*			

- *Output: Preliminary software concepts for near-target and far-target use-cases*

Phase 2 (30 months)

Key Task 1: Refinement of bio-medical JCAS concept based on two concrete use-cases (3 months)

- *Input: Preliminary hardware and software concepts (milestone M1)*
- *Output: JCAS block-level diagram and requirements, input/output specifications for building blocks*

Key Task 2: Bio-medical JCAS radar and communication front-ends (8 months)

- *Input: Milestone 1 from phase 1*
- *Output: Designed, fabricated and experimentally validated JCAS radar and communication front ends*

Key Task 3: Bio-medical JCAS filters (6 months)

- *Input: Milestone 1 from phase 1*
- *Output: Designed, fabricated and experimentally validated JCAS filters*

Key Task 4: Bio-medical JCAS baseband electronics (4 months)

- *Input: Milestone 1 from phase 1*
- *Output: Designed and validated JCAS baseband electronics sub-system*

Key Task 5: Bio-medical JCAS MIMO-antennas (9 months)

- *Input: Milestone 1 from phase 1*
- *Output: Designed, fabricated and experimentally validated JCAS MIMO-antennas sub-system*

Key Task 6: Development and test of bio-medical JCAS software algorithms (9 months)

- *Input: Milestone 1 from phase 1*
- *Output: Tested JCAS software algorithms for radar sensing, wireless communication and AI processing*

Key Task 7: Implementation of bio-medical JCAS software for radar signal processing (3 months)

- *Input: Milestone 1 from phase 1, output from key task 6*
- *Output: JCAS firmware bio-medical JCAS radar signal processing implemented and vital signs extracted*

Key Task 8: Implementation of bio-medical JCAS software for wireless communication (3 months)

- *Input: Milestone 1 from phase 1, output from key task 6*
- *Output: JCAS firmware bio-medical JCAS wireless communication*

Key Task 9: Implementation of AI and machine learning based vital sign analysis (3 months)

- *Input: Milestone 1 from phase 1, output from key task 6*
- *Output: AI-based vital sign analysis for deduction of vital sign abnormalities*

Key Task 10: Bio-medical JCAS demonstrator manufacturing and set-up (5 months)

- *Input: Milestone 2, software components from key tasks 7-9*
- *Output: Validated bio-medical JCAS system demonstrator*

Key Task 11: Bio-medical JCAS demonstration and evaluation (3 months)

- *Input: Milestones 2 and 3*
- *Output: Evaluated bio-medical JCAS system for real-time extraction of vital signs and wireless transmission of measured data*

Milestones for Phases 1 and 2

- ***Milestone Phase 1**: Preliminary hardware and software concepts completed for bio-medical JCAS system*

- ***Milestones Phase 2***
  - *Milestone M2: Validated bio-medical JCAS sub-systems ready for demonstrator integration*
  - *Milestone M3: Validated AI and machine learning based algorithms ready for demonstrator implementation*
  - *Milestone M4: Bio-medical JCAS demonstrator ready for experiments*

4. R&R (Roles and Responsibilities)

Expected R&R of Research Institute	*Fraunhofer* • *Bio-medical JCAS system **hardware** design, manufacturing and testing* • *Integrate **hardware** building-blocks into bio-medical JCAS system* • ***Joint demonstration** and evaluation of bio-medical JCAS system*
Expected R&R of Korean SME Partner	*SME* • *Bio-medical JCAS **software** design, testing and implementation* • *Integrate **software** building-blocks into bio-medical JCAS system* • ***Joint demonstration** and evaluation of bio-medical JCAS system*

5. Expected Outcomes/Impact

Expected Outcomes and Impact	• *Functional bio-medical JCAS system including both hardware and software components* • *Evaluation results of bio-medical JCAS system for mobile and stationary use-cases* • *High potential for commercialization for applications in monitoring of infants, burn victims and people with sleep disorders, to patients in nursing homes, and even drivers and passengers in vehicles, especially children* • *Contribution to telemedicine*

6. References

(1) Contactless Patient Monitoring — ECG using Radar, Fraunhofer Research News, 05 / 2025,

https://www.fraunhofer.de/en/press/research-news/2025/may-2025/contactless-patient-monitoring-ecg-using-radar.html

(2) S. C. Pasupuleti, T. Huyen Le and I. Ndip, "Package Integrated Series-feed Antenna Arrays for mmWave Medical Radar Sensors," 2025 16th German Microwave Conference (GeMiC), Dresden, Germany, 2025, pp. 33-36, doi: 10.23919/GeMiC64734.2025.10979173

(3) Providing care when it is needed – Radar system recognizes and tracks people and objects

Safety: Radar system recognizes and tracks people and objects

☐ Terms:

1) Industrial Technology Classification

· Classification system according to Industrial Technology Classification Table (**Appendix 1**, based on Article 16 of the Common Operational Guidelines for the Industrial Technology Innovation Project in Korea).

2) Technology Readiness Level (TRL)

Description		
·A consistent and objective indicator of the technological maturity of key element technologies		
	Basic Research Stage (1-2)	1st Stage: Basic theories and experiments 2nd Stage: Proof of Concept, such as practical ideas or patents
	Experimental Stage (3-4)	3rd Stage: Verification of basic performance at a laboratory scale 4th Stage: Evaluation of key performance of materials, components, and systems at a laboratory scale
	Prototype Stage (5-6)	5th Stage: Production and performance evaluation of prototypes of finalized materials, components, and systems 6th Stage: Production and performance evaluation of prototypes at a pilot scale
	Productization Stage (7-8)	7th Stage: Reliability testing and evaluation by potential customers 8th Stage: Prototype certification and standardization
	Commercialization Stage (9)	9th Stage: Commercialization